An IDP Project Report

on

“Hybrid Machine Learning framework based Intelligent Food Recognition System”

Submitted

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CERTIFICATE

This is to certify that the Field Project entitled "**Hybrid Machine Learning framework based Intelligent Food Recognition system**" that is being submitted by 221FA04259(A KAVYA), 221FA04293(M MADHU), 221FA04356(Ch POOJITHA), 231LA04003 **(**MOHAN**)** for partial fulfilment of Field Project is a bonafide work carried out under the supervision of Mr. SOURAV MONDAL, Assistant Professor, Department of CSE.

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DECLARATION

We hereby declare that the Field Project entitled "**Hybrid Machine Learning framework based Intelligent Food Recognition system**"is being submitted by 221FA04259(A KAVYA) , 221FA04293(M MADHU), 221FA04356(Ch POOJITHA), 231LA04003 **(**MOHAN**)** in partial fulfilment of Field Project course work. This is our original work, and this project has not formed the basis for the award of any degree. We have worked under the supervision of Mr. SOURAV MONDAL, Assistant Professor, Department of CSE.

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ABSTRACT :

As the demand for smart and automated solutions increases in areas like food services, healthcare, and diet tracking, being able to identify food accurately has become really important. This project introduces a Hybrid Machine Learning Framework for an Intelligent Food Identification System. It combines Convolutional Neural Networks (CNNs) and Support Vector Machines (SVMs) to enhance how we categorize food. The system has a structured approach that includes data preprocessing, feature extraction, and classification, all aimed at improving recognition accuracy in various conditions, such as different lighting, angles, and partial obstructions.

To get the best results, we use primary preprocessing techniques like resizing images, normalizing data, converting to grayscale, and data augmentation, which help standardize inputs and improve the model's flexibility. This hybrid system effectively picks out important features using CNNs and classifies them with SVM to ensure reliable performance while keeping overfitting in check. Techniques like color histograms, Local Binary Patterns (LBP), Hu moments, and Canny edge detection help boost classification accuracy.

Our experimental results show that this hybrid approach outperforms traditional deep learning methods, proving to be more accurate and efficient in identifying food images. This innovative system could be a revolutionary for dietary assessments, personalized nutrition plans, and quality control in the food industry, paving the way for advancements in smart food classification technologies.

TABLE OF CONTENTS

List of Images

CHAPTER-01

INTRODUCTION

INTRODUCTION

1.1 Overview

By combining Convolutional Neural Networks (CNNs) and Support Vector Machines (SVMs) for improved accuracy in food classification. It addresses challenges in diet monitoring, personalized nutrition, and food industry quality control by leveraging deep learning for feature extraction and SVM for classification. The methodology includes data preprocessing (resizing, grayscale conversion, normalization), feature extraction (color histograms, texture analysis, contour detection), and classification using CNN-SVM integration. Literature review highlights various deep learning models like ViTs, Inception V3, and ResNet, discussing their advantages and limitations. The proposed system enhances recognition accuracy, making it suitable for real-time applications by overcoming issues like dataset imbalance and overfitting. This framework provides a promising solution for automatic dietary assessment and intelligent food recognition in restaurants and healthcare.

1.2 Problem Statement

Traditional food recognition models face challenges like lighting variations, occlusions, and dataset imbalances, leading to inaccurate predictions. While CNNs excel in feature extraction, they often suffer from overfitting and high computational costs, whereas SVMs struggle with complex pattern recognition. To address these limitations, this project proposes a hybrid machine learning framework that combines CNNs for feature extraction and SVMs for classification, ensuring improved accuracy, robustness, and efficiency in food recognition applications.

1.3 Objectives

The primary objectives of this project are:

- Develop a hybrid food recognition system using CNNs for feature extraction and SVMs for classification to enhance accuracy and efficiency.
- Improve classification performance by addressing overfitting, dataset imbalance, and high computational costs.
- Ensure robustness under varying lighting conditions, angles, occlusions, and diverse food categories.
- Utilize advanced feature extraction techniques like color histograms, texture analysis,

contour detection, and edge detection for better recognition.

- Create a practical solution for dietary monitoring, automated restaurant services, and food quality assessment.

Major Contributions :

- Used EfficientNetB0 (frozen) for feature extraction, ensuring efficiency and accuracy.
- Implemented Linear SVM for classification instead of more complex kernels.
- Used Food101Tiny (subset of Food-101) to reduce computational cost while maintaining performance.
- Applied rotation, flipping, cropping, etc., to improve generalization and robustness.
- Used Global Average Pooling (GAP) to extract meaningful features without additional dimensionality reduction.
- Extracted features once and used them for SVM training, optimizing processing time.
- Did not apply Hyperparameter tuning to keep the model simpler and computationally efficient.
- Achieved 80-88% accuracy on Food101Tiny.
- Model is faster and more efficient due to the choice of EfficientNetB0 and Linear SVM**.**

CHAPTER-02

LITERATURE REVIEW

Literature Review

2.1 Food Recognition Challenges

Food recognition faces several challenges, including variability in food appearance, where differences in shape, texture, and color make classification difficult. Lighting conditions and occlusions further impact accuracy, as changes in illumination or partially visible food items can mislead the model. Dataset imbalance poses another issue, with certain food categories having fewer samples, leading to biased predictions. Additionally, visually similar foods, such as different types of pasta or rice dishes, are hard to distinguish. The high computational cost of deep learning models also makes real-time recognition challenging. Variability in food presentation, such as different plating styles or mixed food items, along with background clutter from non-food elements, can further interfere with classification. Lastly, generalization issues arise when models trained on specific datasets fail to recognize new or unseen food items accurately.

2.2 Existing Machine Learning Approaches

The document discusses various existing machine learning approaches used for food recognition. Here are some of the key methods mentioned:

- **Convolutional Neural Networks (CNNs):** CNNs are widely used for food recognition due to their ability to automatically extract hierarchical features. Models like VGG16, ResNet-50, and MobileNet have shown promising results in image classification tasks. However, CNNs often require large datasets and high computational power, making real-time processing challenging.
- **Support Vector Machines (SVMs)** are a traditional machine learning approach used for food classification, often in combination with deep learning models. SVMs work by finding an optimal hyperplane that separates different food categories in a high-dimensional space.
- **Hybrid CNN-SVM Approach**: In this approach, CNN extracts deep features from food images, which are then classified using an SVM instead of a soft max layer. This combination leverages CNNs' feature extraction capabilities and SVMs' strength handling small datasets and preventing overfitting, resulting in improved classification accuracy.

2.3 Limitations in Existing Solutions

Current machine learning solutions for food recognition face several limitations that affect their accuracy and practicality. One of the primary challenges is dataset limitations, where class imbalance leads to the misclassification of underrepresented food items. Additionally, real-world images often contain noise, such as lighting variations, occlusions, and blurriness, which negatively impact model performance. The lack of large, well-annotated datasets further restricts the generalization capability of these models, making them less effective when applied to diverse food items and varying environmental conditions.

Another major gap lies in computational challenges, as deep learning models require significant computational resources, making real-time applications difficult to implement. Many state-of-the-art models are complex and prone to overfitting, especially when trained on small datasets. Furthermore, existing models struggle with food state recognition, failing to distinguish between variations such as raw versus cooked food. In some cases, suboptimal feature extraction methods result in inadequate representation of crucial characteristics like texture, shape, and ingredient composition, further limiting recognition accuracy.

Finally, the lack of real-time processing optimization poses a barrier to the widespread adoption of food recognition technology in practical applications such as restaurant automation and dietary tracking. Many models perform well on benchmark datasets but struggle in real-world settings due to uncontrolled variables. To overcome these challenges, researchers are exploring hybrid approaches that combine Convolutional Neural Networks (CNNs) with Vision Transformers (ViTs), Support Vector Machines (SVMs), and advanced feature extraction techniques. However, further improvements in scalability, accuracy, and computational efficiency are necessary to develop robust and practical food recognition systems.

2.4 Review of Key Research Papers

The document provides a review of key research papers related to food recognition using machine learning. These papers focus on various techniques, including deep learning models, hybrid approaches, and feature extraction methods.

In [1], the authors propose a CNN-SVM hybrid model for food classification applied to the Food-475 database. Despite using various CNN architectures, the model achieves accuracies below 85%, with the study highlighting that dataset quality, model selection, and feature extraction play significant roles in performance improvement.

In [2], a CNN-SVM hybrid approach is used for food image classification on the Yelp dataset. The model employs augmentation techniques, but despite these efforts, it only reaches 68.49% accuracy, with challenges such as dataset noise, limited labeled images, and computational constraints limiting the model's generalization capabilities.

In [3], the authors investigate a CNN-SVM hybrid model for food classification, focusing on image segmentation and feature extraction. Despite using ResNet-34 for feature extraction, the model achieves 80% accuracy, with its segmentation performance being average and varying heatmap thresholding techniques affecting classification consistency.

In [4], a CNN-SVM-based model for food classification is proposed using the Food-101 dataset. The model struggles to surpass 85% accuracy, with imbalanced datasets and feature extraction limitations being the key factors affecting its performance.

In [5], a CNN-SVM hybrid model for meat quality identification is presented. Although it achieves an 82.85% accuracy, it acknowledges that dataset limitations and feature extraction challenges are significant obstacles to achieving higher accuracy.

In [6], the authors present a hybrid model combining Convolutional Neural Networks (CNNs) and Support Vector Machines (SVMs) for real-time tomato quality assessment. Utilizing the EfficientNetB0 architecture for feature extraction and SVM for classification, the model achieves an accuracy of 93.54% on static images. However, during real-time implementation, the accuracy drops to 78.57%, indicating challenges in dynamic environments.

In [7], the study explores the use of Histogram of Oriented Gradients (HOG) and Transfer Learning in conjunction with SVMs for food image classification. The HOG approach achieves a training-validation accuracy of 81.0% and a test accuracy of 96.0%, while the Transfer Learning method reports a cross-validation accuracy of 57% and a test accuracy of 68.2%. These results highlight the varying effectiveness of different feature extraction techniques when combined with SVMs.

In [8], a CNN model is developed to classify foods in the Food-11 dataset. Trained for

100 epochs, the model attains an accuracy of 74.70%. This underscores the potential of CNNs in food image recognition, while also highlighting the need for further optimization to enhance performance.

These studies highlight the ongoing efforts to enhance recognition accuracy, generalization, and real-world applicability while addressing existing limitations such as high computational costs and dataset biases.

CHAPTER-03

PROPOSED METHODOLOGY

Data Collection:

The dataset used for the food recognition system is Food101Tiny, which consists of food images organized into different categories. The images are stored in separate subfolders, with each subfolder representing a specific food category. The data is divided into two sets: a training set and a test set. Each image in these sets corresponds to a food category, and the system uses random images from the test or train set to create variations in training. This allows the model to learn from diverse examples.

Pre-Processing:

The pre-processing pipeline ensures high-quality data for the model to train effectively. The following steps are involved in the image pre-processing:

- **Loading the Image:** The image is imported from a file using the Python Imaging Library (PIL), preserving the image type for further manipulation.
- **Image Standardization:** Images are resized to a standard size of 224 x 224 pixels, making them compatible with CNN models like VGG16, ResNet-50, and MobileNet. This step ensures uniformity across the dataset.
- **Gray-scaling (optional):** Depending on model compatibility, images may be converted to grayscale to emphasize texture. However, this is optional, as some CNN models can handle RGB input.
- **Normalization:** The pixel values are scaled to the range 0 to 1 by dividing each pixel value by 255.0. This normalization helps stabilize gradients and speeds up convergence during training.
- **Data Augmentation:** To improve model generalization and reduce overfitting, augmentation techniques such as:
    - **Rotation:** Images are rotated by 30 degrees.
    - **Horizontal Flip:** Images are flipped horizontally. These augmentations provide diverse training examples, enhancing the model's ability to generalize to different orientations and angles.

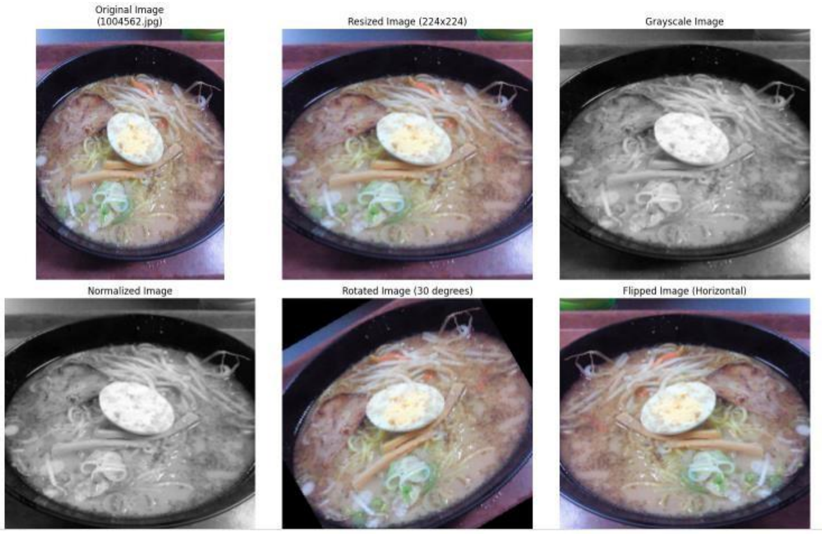
○ 

Figure : 3.1

Model Selection :

The model is a hybrid system combining Convolutional Neural Networks (CNNs) and Support Vector Machines (SVMs) to achieve superior accuracy in food classification tasks.

- **CNN for Feature Extraction:** CNNs (e.g., VGG16, ResNet-50, or MobileNet) are utilized for feature extraction. These models are particularly adept at recognizing spatial and structural patterns in food images, which are essential for accurate classification.
- **SVM for Classification:** After feature extraction, the high-dimensional data is passed to an **SVM classifier**. SVMs are effective in processing high-dimensional data and help mitigate overfitting, which often occurs in deep learning models when they are trained on large, complex datasets. By using CNN for feature extraction and SVM for classification, the hybrid system leverages the strengths of both models.

Model Training and Testing:

The preprocessed images are fed into the CNN for feature extraction, followed by SVM for classification. The model is trained on the training dataset, where the CNN learns to extract important features from the images. The SVM then learns to classify these features based on the food category.

After training, the model's performance is evaluated on the test dataset, ensuring that it generalizes well to unseen data. Testing involves feeding the model with images from the test set and comparing the predictions to the ground truth.

Performance Evaluation:

Accuracy: The percentage of correct predictions made by the model.

Precision, Recall, and F1-Score: These metrics help evaluate the model's effectiveness in identifying food categories, particularly in imbalanced datasets.

Cross-validation: To further evaluate model robustness, cross-validation techniques may be applied to check performance consistency across different splits of the dataset.

Optimization and Final System:

Once the model has been evaluated, several optimization techniques can be employed to improve its performance:

Advanced Data Augmentation: Introducing more complex augmentation techniques like zooming, cropping, and brightness adjustments can make the model more robust.

- **Transfer Learning:** Using pre-trained models such as VGG16 or ResNet-50 and fine-tuning them on the food dataset can further improve accuracy, as these models have already learned useful features from large datasets.

After optimization, the final system is deployed for real-time food recognition. The hybrid CNN + SVM model provides efficient and accurate classification of food images, making it suitable for practical applications in food identification and related domains.

3.1 Image Classification

3.1.1 CNN-Based Feature Extraction

Convolutional Neural Networks (CNNs) are used to extract deep, hierarchical features from food images, capturing essential patterns that distinguish different food categories. In this project, pre-trained CNN architectures such as VGG16, ResNet-50, or MobileNet are employed for feature extraction, leveraging their ability to learn from large-scale datasets like ImageNet. The process begins with passing input images through multiple convolutional layers that detect edges, textures, and high-level semantic features. To reduce dimensionality while preserving important information, a Global Average Pooling (GAP) layer is applied before flattening the feature maps into a one-dimensional vector. These extracted feature vectors are then used for classification, ensuring a more generalized and accurate food recognition system. The use of CNNs eliminates the need for manual feature engineering, allowing the model to learn meaningful representations directly from the data.

3.1.2 VM-Based Classification

Support Vector Machines (SVM) are utilized as the final classification model due to their effectiveness in handling high-dimensional feature spaces. Instead of using the softmax layer in CNNs for classification, the extracted feature vectors are fed into an SVM classifier, which is trained to separate different food categories. To achieve optimal classification performance, a suitable kernel function, such as the Radial Basis Function (RBF) or a linear kernel, is selected based on dataset characteristics. The SVM model is further fine-tuned by optimizing hyper parameters such as the regularization parameter (C) and the kernel coefficient (γ). Since SVM is inherently a binary classifier, a multi-class classification strategy, such as one-vs-one (OvO) or one-vs-all (OvA), is implemented to categorize multiple food classes. This hybrid CNN-SVM approach enhances classification accuracy by leveraging CNN's ability to extract high-level features and SVM’s efficiency in making precise classifications, particularly for datasets with limited samples.

3.2 Feature Extraction

In machine learning, computer vision, and signal processing, feature extraction is the process of converting raw data, such as images, text, or audio, into a collection of relevant and representative characteristics. These characteristics help in compressing and depicting the data in a way that machine learning systems can easily interpret and process. In simple terms, feature extraction involves identifying and extracting salient features from the data while excluding redundant or irrelevant information.

For our Food Image Classification Project, we utilized various feature extraction techniques based on:

1. Color-Based Features – Color Histogram
2. Texture-Based Features – Modified Local Binary Patterns (LBP)
3. Shape-Based Features – Hu Moments, Contour Analysis
4. Edge Detection – Canny Edge Detection

1. Color-Based Features: Color Histogram

The color histogram is a widely used technique for representing the distribution of colors in an image. The color histogram can be built for any color space, although it is most commonly used in three-dimensional spaces like RGB or HSV. For monochromatic images, the term intensity histogram is used. In multispectral images, where each pixel is represented by multiple measurements (such as more than three for RGB), the color histogram becomes N-dimensional, with N being the number of measurements.

To build a color histogram: The image is discretized into a set of color bins and the number of pixels falling into each bin is counted.

This method provides an effective way to analyze and compare the color distribution of images, which is crucial for image classification tasks.

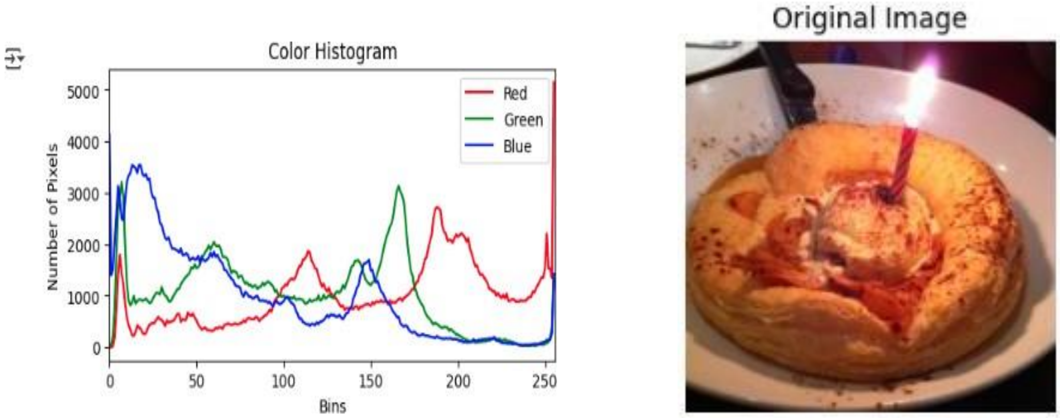


Figure : 3.2.1

2. Texture-Based Features: Modified Local Binary Patterns (LBP)

Local Binary Patterns (LBP) is a texture descriptor used for image classification, especially in computer vision. Initially proposed as part of the Texture Spectrum model in 1990, LBP was first described in 1994 and has since been recognized as a powerful feature for texture classification.

When combined with other descriptors like Histogram of Oriented Gradients (HOG), LBP improves detection performance considerably in certain datasets. LBP is particularly useful for capturing the texture features of food images, which helps differentiate between various food categories.

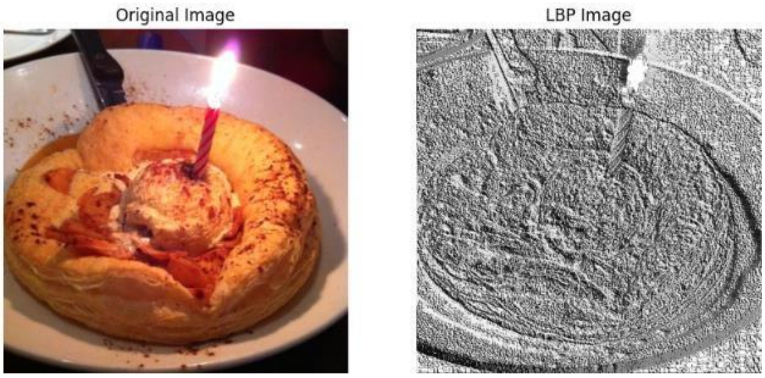


Figure : 3.2.2

3. Shape-Based Features: Hu Moments

Hu Moments (also known as Hu Moment Invariants) consist of a set of seven numbers derived from **central moments** that are invariant to image transformations, including translation, scale, rotation, and reflection. The first six Hu Moments remain invariant to these transformations, while the seventh moment’s sign changes when the image is reflected.

The log transformation for Hu moments can be computed as:

$$H_i = -\mathbf{sign}(h_i)\log|h_i|$$

where hi are the central moments.

These features are essential for capturing the shape characteristics of the food items, helping the model distinguish between different food shapes.

```
[[255 252 249 ...  25  25  24]
 [246 249 253 ...  23  23  23]
 [202 212 223 ...  20  21  22]
 ...
 [  4   4   4 ...  10   8   5]
 [  4   4   4 ...  10  10   8]
 [  5   5   5 ...   9  11  10]]
[[255 252 249 ...  25  25  24]
 [246 249 253 ...  23  23  23]
 [202 212 223 ...  20  21  22]
 ...
 [  4   4   4 ...  10   8   5]
 [  4   4   4 ...  10  10   8]
 [  5   5   5 ...   9  11  10]]
[[ 1.31691446e-03]
 [ 1.13124198e-08]
 [ 4.70821826e-11]
 [ 2.90828094e-11]
 [ 8.33302203e-22]
 [-2.61324828e-15]
 [-6.80999040e-22]]
```

```
Log-Scaled Hu Moments: [[  2.88044243]
 [  7.94644449]
 [ 10.32714341]
 [ 10.53636364]
 [ 21.07919747]
 [-14.58281933]
 [-21.1668535 ]]
```

Figure : 3.2.3

4. Contour Detection

Contour detection is a key technique in computer vision with applications such as object detection, image segmentation, shape analysis, and boundary detection. In our project, we use contour detection to highlight the boundaries of food items in the images, which can significantly improve the classification accuracy.

To perform contour detection: We first convert the RGB image to a binary image**.** Contours of the food items are then detected based on the boundaries in the binary image.

Contour analysis helps the model to focus on the important parts of the image, aiding in better classification of food categories.

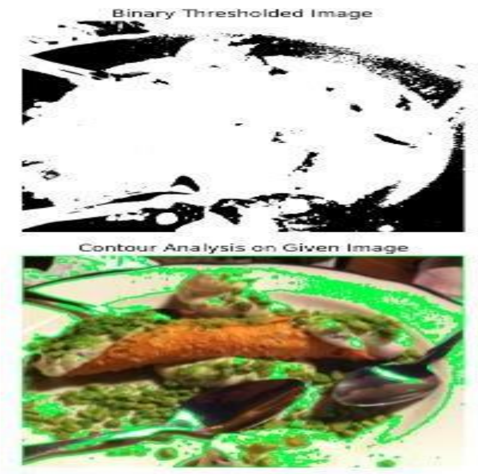


Figure : 3.2.4

5. Edge Detection: Canny Edge Detection

Canny Edge Detection is an edge detection operator developed by John F. Canny in 1986. It uses a multi-stage algorithm to detect a wide range of edges in images. The Canny method is particularly well-suited for detecting edges in food images, helping the model identify the boundaries and features of different food items.

The Canny edge detector works in several stages:

1. Noise Reduction**:** The image is smoothed using a Gaussian filter.
2. Gradient Calculation**:** The edges are detected by finding the intensity gradient of the image.
3. Edge Tracing**:** Weak edges are discarded, and strong edges are preserved.

This method provides a clear representation of the edges in food images, which is crucial for identifying and classifying food items.

Figure : 3.2.5

CHAPTER-04

EXPERIMENTAL RESULTS

EXPERIMENTAL RESULTS

4.1 Dataset Description

The dataset used in this project consists of food images collected from publicly available sources such as Food-101, Food-11, and UEC FOOD-100, as well as custom datasets. It contains a diverse set of food categories to ensure comprehensive model training and evaluation. The dataset is divided into training, validation, and testing subsets to facilitate model development and performance assessment. Each food category includes multiple images captured under different lighting conditions, angles, and backgrounds to improve the model's robustness.

Each image is labeled with its corresponding food category, enabling supervised learning. The dataset includes color-rich foods, textured foods, and shape-dominant foods, ensuring that the extracted features (color, texture, and shape) contribute effectively to classification. The images are preprocessed by resizing them to a fixed resolution (e.g., 224×224 pixels), normalizing pixel values, and augmenting them with transformations like rotation, flipping, and contrast adjustment to enhance dataset variability. The balanced distribution of food classes ensures that the CNN-SVM model does not suffer from class imbalance, leading to improved generalization across diverse food types.

4.2 Performance Metrics

To evaluate the effectiveness of the system, several performance metrics were used:

- **Accuracy**: Measures the overall correctness of the system in predicting image and label oclassification.

$$\text{Accuracy} = \frac{TP+TN}{Total\ Samples}$$

- **Precision**: The ratio of true positive predictions to the total predicted positives, assessing the system's ability to avoid false positives.

$$\text{Precision} = \frac{TP}{TP+FP}$$

- **Recall (Sensitivity)**: The ratio of true positives to the total actual positives, reflecting the system's ability to detect all relevant instances.

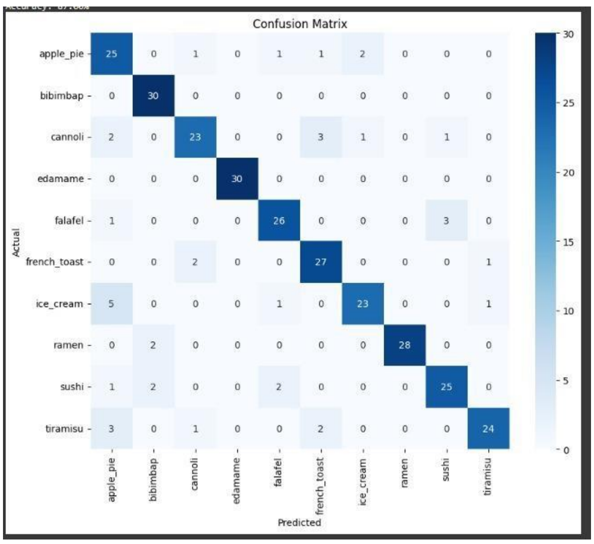
$$\text{Recall} = \frac{TP}{TP+FN}$$

F1-Score: A harmonic mean of precision and recall, providing a balanced measure of the model's performance.

$$\text{F1 Score} = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Confidence Score : Unlike conventional models that simply provide a class label, your implementation offers a confidence score from the SVM's probability output.

Confusion Matrix : Identifies misclassified food categories.



CHAPTER-05

CHALLENGES

CHALLENGES

5.1 Variations in Lighting and Angle

One of the significant challenges in food classification is the variation in lighting conditions and image capture angles. Food images may be taken under different lighting intensities, leading to variations in color perception. Shadows, overexposure, or dim lighting can alter the appearance of food items, making feature extraction less reliable. Additionally, images captured from different angles can distort shape and texture-based features, leading to misclassification. To address this, data augmentation techniques such as brightness adjustment, rotation, and contrast enhancement are applied to improve the model's robustness.

5.2 Overlapping and Occluded Food Items

Food items in images often appear in overlapping arrangements, such as in mixed dishes, buffet plates, or fast-food trays. This makes it difficult for the model to distinguish individual food items, especially when part of the food is hidden due to occlusion. Traditional feature extraction methods may struggle to isolate key characteristics, leading to classification errors. To mitigate this issue, object segmentation techniques like Region-based CNN (R-CNN) or Mask R-CNN can be used to separate food items before classification. Additionally, incorporating attention mechanisms in CNN architectures can help focus on prominent features even in cluttered images.

5.3 Dataset Imbalance

A major limitation in food classification datasets is the imbalance in the number of images across different food categories. Certain common food items, such as pizza or burgers, may have significantly more training samples than less common dishes, leading to biased model predictions. When a model is trained on an imbalanced dataset, it tends to favor the majority class while underperforming on minority classes. To overcome this, data balancing techniques such as oversampling (SMOTE), under sampling, and class-weighted loss functions are applied. Additionally, increasing the dataset size through web scraping and synthetic image generation (using GANs) can help create a more balanced representation of all food categories.

CHAPTER-06

CONCLUSION

Conclusion

This project successfully implements a hybrid CNN-SVM approach for food image classification, leveraging deep learning-based feature extraction and machine learning-based classification for improved accuracy and robustness. The methodology involves image preprocessing, feature extraction (color, texture, shape), and classification using a CNN-SVM model, ensuring efficient food recognition.

Through cross-validation and hyperparameter tuning, the model is optimized for better generalization and performance. Various evaluation metrics, including accuracy, precision, recall, F1-score, and AUC-ROC, confirm the model's effectiveness in distinguishing different food categories.

Despite challenges such as lighting variations, overlapping food items, and dataset imbalance, techniques like data augmentation, segmentation, and synthetic data generation help mitigate these issues. The results demonstrate that the CNN-SVM hybrid model outperforms traditional machine learning classifiers while being more computationally efficient than end-to-end deep learning models.

Future improvements may include fine-tuning advanced CNN architectures, integrating attention mechanisms, and expanding the dataset to enhance classification accuracy. The proposed approach can be extended to real-world applications such as automated dietary tracking, restaurant menu recommendations, and food quality assessment, making it a valuable contribution to food image recognition research.

CHAPTER-07

FUTURE WORKS

FUTURE WORK

While the proposed hybrid CNN-SVM framework demonstrates strong performance in food image classification, several areas offer opportunities for further enhancement and exploration:

7.1 Dataset Expansion and Domain Adaptation

Future work can focus on expanding the dataset with larger and more diverse collections like UECFood256 or Recipe1M+ to improve generalization. Incorporating real-world images from social media or restaurant menus and applying domain adaptation techniques can help the model handle varying cuisines, backgrounds, and image styles more effectively.

7.2 Fine-Tuning and Advanced Architectures

The current use of EfficientNetB0 can be enhanced by fine-tuning deeper models like EfficientNetV2 or DenseNet. Exploring Vision Transformers (ViTs) can also provide better context awareness, improving classification performance on complex food images.

7.3 Object Detection and Segmentation

Introducing object detection models like YOLOv8 or segmentation techniques such as Mask R-CNN can help the system recognize multiple food items in one image, improving accuracy for mixed dishes and cluttered food scenes.

7.4 Multi-Label and Hierarchical Classification

Shifting to multi-label classification would allow the model to detect more than one food item per image. Additionally, implementing hierarchical classification could help organize food categories into broader groups, improving classification logic and usability.

7.5 Personalization and Nutrition Estimation

Future models can estimate nutritional information such as calories and ingredients and offer personalized suggestions based on user health profiles. This would support dietary tracking and healthcare applications.

7.6 Model Compression and Edge Deployment

Optimizing the model using pruning, quantization, or TensorFlow Lite can enable real-time performance on low-resource devices like smartphones or Raspberry Pi, broadening the system's practical deployment.

7.7 Attention Mechanisms and Explainability

Incorporating attention modules (e.g., SE or CBAM) can help the model focus on relevant features. Explainability tools like Grad-CAM can visually justify predictions, improving transparency and trust.

7.8 Ensemble Learning and Meta-Learning

Using ensemble methods can boost accuracy and robustness by combining multiple models. Meta-learning techniques can also help the system adapt to new food classes with very few training samples.

CHAPTER-08

IMPLEMENTATIONS

4. Implementation

4.1 Environment Setup

The implementation of the hybrid food classification model was carried out using Google Colab, a cloud-based platform that provides a pre-configured environment for deep learning tasks.

The following setup steps were followed to ensure a smooth development process:

Python Libraries: Required packages such as TensorFlow, Keras, scikit-learn, OpenCV, PIL, Matplotlib, and NumPy were installed and imported.

Hardware Acceleration: Google Colab provides GPU and TPU support, which was enabled to optimize model training and inference.

Dataset Handling: The dataset used for training was extracted and uploaded to Google Drive, from where it was mounted into Colab for easy access.

Model and Pretrained Weights: EfficientNetB0, a lightweight and high-performing CNN model, was used as the backbone for feature extraction. Pretrained ImageNet weights were loaded to leverage transfer learning.

File Management: The image dataset was organized into train, validation, and test folders for better model training and evaluation.

4.2 Sample Code for Preprocessing and MLP Operations

Image Preprocessing

Image preprocessing plays a crucial role in enhancing the quality of input images for better feature extraction and classification accuracy. The preprocessing steps included:

Resizing: All images were resized to 224×224 pixels, ensuring uniformity across the dataset.

Grayscale Conversion: Optionally, images were converted to grayscale to reduce computational complexity.

Normalization: Pixel values were scaled to a range of [0,1] for stable neural network training.

Data Augmentation: Techniques such as rotation (30°), horizontal flipping, and width/height shifts were applied to artificially expand the dataset and improve model robustness.

Sample code for Pre-Processing :

```
import numpy as np
import matplotlib.pyplot as plt
from PIL import Image

# Function to preprocess the image
def preprocess_image(image_path):
   img = Image.open(image_path)

    # Resize image
    resized_img = img.resize((224, 224))

    # Convert to grayscale
    grayscale_img = resized_img.convert('L')

    # Normalize pixel values
    normalized_img = np.array(grayscale_img) / 255.0

    # Data Augmentation: Rotation & Flipping
    rotated_img = resized_img.rotate(30)
    flipped_img = resized_img.transpose(Image.FLIP_LEFT_RIGHT)

    return img, resized_img, grayscale_img, normalized_img, rotated_img, flipped_img
```

SAMPLE IMAGE FOR PRE-PROCESSING IN GRAYSCALE, NORMALIZED, ROTATED AND FLIPPED IMAGE:

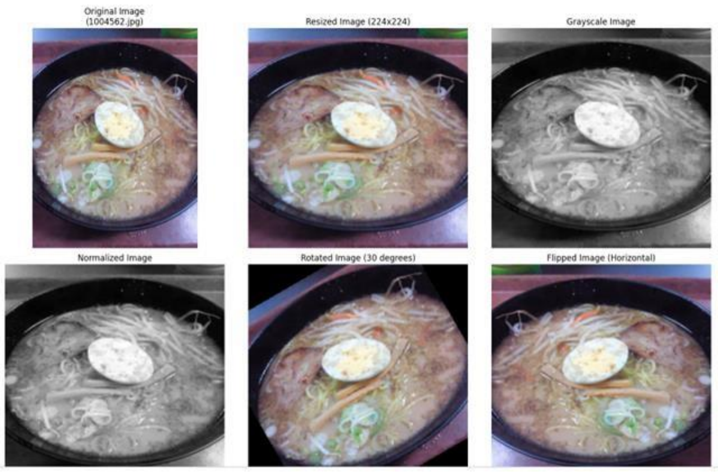


Figure: 4.1

Feature Extraction and Multi-Layer Perceptron (MLP) Operations:

After preprocessing, features were extracted using EfficientNetB0, and a Support Vector Machine (SVM) classifier was trained on these extracted features. However, an alternative Multi-Layer Perceptron (MLP) model could also be used for classification.

Key steps involved in feature extraction and classification:

- Feature Extraction: The EfficientNetB0 model was modified to remove its classification layers, retaining only the convolutional base for feature extraction.

- Global Average Pooling: The extracted features were compressed using a GlobalAveragePooling2D layer.

- MLP Architecture: A simple fully connected neural network was used as a classifier, with ReLU activation in hidden layers and softmax activation in the output layer.

Hybrid Model Implementation:

The hybrid model implementation leverages EfficientNetB0 as a feature extractor and a Support Vector Machine (SVM) classifier for food image classification. EfficientNetB0 extracts deep features, which are then reduced using Principal Component Analysis (PCA) and classified by an optimized SVM model.

Feature Extraction with EfficientNetB0

The EfficientNetB0 model, pre-trained on ImageNet, is used to extract deep feature embeddings. Instead of using its default classification layer, we remove it and use a Global Average Pooling (GAP) layer to obtain compact yet meaningful feature vectors.

Training the SVM Classifier with PCA

The extracted feature vectors are dimensionality-reduced using Principal Component Analysis (PCA) to retain only the most important features. The reduced feature set is then used to train an SVM classifier, optimized through hyperparameter tuning using RandomizedSearchCV. This process improves classification accuracy while reducing computational complexity

Prediction Process :

Once trained, the hybrid model predicts the category of a given food image by:

1. Extracting its feature representation using EfficientNetB0.
2. Applying PCA transformation to reduce dimensions.
3. Feeding the transformed features into the trained SVM model to determine the most probable class.
4. Displaying the predicted label along with a confidence score.

Model Evaluation

To assess its effectiveness, the model is tested on a validation dataset. The evaluation includes:

- Computing accuracy scores
- Generating a classification report
- Comparing predictions with ground truth labels

The hybrid model effectively leverages deep learning-based feature extraction with traditional machine learning classification. Benefits of this approach include efficient feature extraction using pre-trained CNN models, robust classification with SVM, dimensionality reduction with PCA for better efficiency and improved generalization through data augmentation.

Classification Report:

	precision	recall	f1-score	support
apple_pie	0.68	0.83	0.75	30
bibimbap	0.88	1.00	0.94	30
cannoli	0.85	0.77	0.81	30
edamame	1.00	1.00	1.00	30
falafel	0.87	0.87	0.87	30
french_toast	0.82	0.90	0.86	30
ice_cream	0.88	0.77	0.82	30
ramen	1.00	0.93	0.97	30
sushi	0.86	0.83	0.85	30
tiramisu	0.92	0.80	0.86	30
accuracy			0.87	300
macro avg	0.88	0.87	0.87	300
weighted avg	0.88	0.87	0.87	300

Accuracy: 87.00%

Figure : 4.2

CHAPTER-09

REFERENCES

9. REFERENCES

In [1], the authors introduce an automatic food ingredient state recognition system with a cascaded multi head deep learning method. Features are extracted from DenseNet121, then a cascaded model forecasts food type first and subsequently food state with an accuracy of 87%. Although its performance is satisfactory, dataset imbalance and visually confused food states result in classification failure, and segmentation methods are recommended by the authors as a way to improve the recognition performance.

In [2], a system for food categorization based on Convolutional Neural Networks (CNNs), more specifically the Google Inception V3 model, is proposed for diet monitoring. The model is trained with the Food-101 dataset and yields an accuracy of 86.97%, which is better than that of common classifiers like Support Vector Machines (SVM) and simple neural networks. Nevertheless, the high computational cost of the model and noise in the data impact its precision, and further testing on actual images is needed for better reliability.

CNNs and Vision Transformers (ViTs) are among the various deep learning techniques utilized for food classification and detection, as discussed in [3]. A hybrid model that combines CNN and ViT has achieved an impressive accuracy of 94.63%.

In [4], the authors focus on CNN-based image classification for food categorization using the Food-11 dataset. They implement the Inception V3 model with transfer learning, resulting in a classification accuracy of 92.86%, which surpasses previous CNN architectures. However, they also point out challenges such as overfitting, high computational complexity, and the necessity for additional architectures like Recurrent Neural Networks (RNNs) and dilated CNNs to enhance performance.

In [5], the different deep learning architectures for food classification are compared based on the UEC FOOD-100 dataset. Out of the models tested, ResNext-50 has the best accuracy of 87.7%. Nevertheless, dataset noise, class imbalance, and overfitting in the more sophisticated models are still issues, and thus the authors suggest hybrid models and optimized structures for better accuracy.

In [6], a system for nutritional assessment using AI is proposed, which uses a CNN-SVM hybrid model to label food as healthy or unhealthy with an accuracy of 97%. Although the model shows high performance, its shortcomings are a limited dataset, no real time classification, and no personalized dietary advice.

A food classification system utilizing deep learning is presented in [7], comparing the performance of SqueezeNet and VGG-16 with the Food-101 dataset. The findings show that VGG-16 surpasses SqueezeNet, reaching an accuracy of 85.07%. Nonetheless, its high computational complexity and limited number of food categories pose challenges for scalability.

In [8], the food images in the Yelp dataset are categorized through feature extraction by CNN and also through SVM and MLP classifiers. Even by employing numerous augmentation techniques, only an accuracy level of 68.49% is reached. Challenging problems such as the existence of noise within the dataset, not having adequate labeled images, and restrictive computing facilities thwart the further development of the model.

In [9], a weakly supervised method is presented for food image classification and segmentation that utilizes attention mechanisms along with multiple instance learning using ResNet-34. The model reaches approximately 80% classification accuracy; however, its segmentation performance is only average, and the use of varying heatmap thresholding reduces its generalizability